

NOTE · Binary Scaling of ρ

BINARY SCALING OF $\rho = 400/11$

The gap scaling ratio under doubling reveals the framework's prime hierarchy

D. B. – Gap Geometry (updated 5 May 2026; 16 September 2026)
Companion to: Gap Scaling Formula (Paper 4, DOI 10.17605/OSF.IO/C4GK5)
<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>
OSF: osf.io/zx4g7

■ READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.
How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>
DOI 10.17605/OSF.IO/UJWC7

★ UPDATE NOTICE – 16 September 2026 ★

This version adds three lines under the title (a pointer to One coordinate up, where the corpus is now read); nothing else changes.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.
- (2) This paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "Binary Scaling of ρ ".

Every integer claim re-computed 2026-09-11 in exact rationals (the drift sequence to $k = 14$, $5957 = 7 \times 23 \times 37$, the ten residues of the cycle modulo 11, the five multiplicative orders, the seven integer floors and their factorizations, $37 - 400/11 = 7/11$, $36 \div 4/11 = 99$): all stand. Prior version remains on OSF in the project's archive/ folder, named by its date.

★ UPDATE NOTICE – 5 May 2026 ★

This version aligns the §8 evidence-level labels with the canonical methodology taxonomy. The change is definitional. It does not change the scope, honesty, or calculations of this document. Every drift value, factorization, primitive-root order, cycle period, and falsification criterion is unchanged.

Earlier labels: "Level 1 – Exact arithmetic", "Level 2 – Structural observation", "Level 3 – Not claimed". Aligned to the canonical methodology page ([gap-geometry.github.io methodology](https://gap-geometry.github.io/methodology)):
Level 1 – Exact [Theorem], Level 2 – Empirical [Observation],
Level 3 – Conjectural [Conjecture]. The body content under each level is preserved (Level 3 still notes that no Level 3 claim is made in this document).

Prior published version archived as
OLD-2026-05-05_Binary_Scaling_Standalone.txt and
OLD-2026-05-05_Binary_Scaling_Standalone.pdf in OLD/.

This version corrects the Landauer-crossing framing in §2. The earlier text identified step 36 as "the Landauer step in the binary tower (step 36, where $36G \sim 1/\sqrt{2}$).". Both claims in that parenthetical require refinement:

- (a) The Landauer crossing proper is at $n = \ln(2)/G \approx 35.11$ (between steps 35 and 36), not step 36.
- (b) The near-equality " $n \cdot G \approx 1/\sqrt{2}$ " is the GEOMETRIC DAMPING threshold, at $n = 1/(\sqrt{2} \cdot G) \approx 35.82$, not the Landauer threshold.

The refined picture identifies THREE distinct thresholds in the 35–37 zone:

$n \cdot G = \ln 2$	at $n \approx 35.11$	(Landauer erasure cost – the crossing)
$n \cdot G = 1/\sqrt{2}$	at $n \approx 35.82$	(geometric damping)
$n \cdot G = 1/(2 \ln 2)$	at $n \approx 36.54$	(Shannon bound)

The integer $36 = \text{floor}(400/11)$ sits just above the first two thresholds and just below the third. The arithmetic content of §2 – the self-encoding decimal, the 99-closure, the 36.00° pentagonal angle, and the $7/11$ gap to 37 – is unchanged.

Prior published versions remain on OSF file storage, renamed with the OLD- prefix and a date for traceability.

1. THE GAP SCALING RATIO

The gap scaling formula connects the geometric gap $G = 1 - K_{\text{AUD}}$ to the Feigenbaum constant δ :

$$\rho = \delta * G / (\delta - 14/3)$$

where $K_{\text{AUD}} = \sqrt{2} * \ln(2)$ and $\delta = 4.669201609\dots$ is the universal period-doubling constant. This yields:

$$\rho = 400/11 - 1/2500 - 1/939939$$

with agreement to $4 * 10^{-14}$ (Paper 4).

The leading term $400/11 = 36.363636\dots$ carries the formula's structure. This document asks one question: what happens when $400/11$ is scaled by powers of 2 – the framework's native operation?

2. SELF-ENCODING DECIMAL

$$400/11 = 36.363636\dots$$

The repeating block is 36. The integer part is 36. The number contains itself: its decimal expansion echoes its own integer part.

Fractional part: $4/11 = 0.363636\dots$

Integer part: 36

The ratio between them is $36 / (4/11) = 99$. The self-reference closes through $99 = 9 * 11 = (3^2)(11)$.

$$\text{floor}(400/11) = 36 = 6^2 = 2^2 * 3^2$$

This integer sits inside the 35–37 threshold zone of the binary tower. Three distinct thresholds cluster in this zone:

Landauer crossing:	$n \cdot G = \ln 2$	at $n \approx 35.11$
geometric damping:	$n \cdot G = 1/\sqrt{2}$	at $n \approx 35.82$
Shannon bound:	$n \cdot G = 1/(2 \ln 2)$	at $n \approx 36.54$

Step 36 itself is adjacent to the geometric damping threshold: $36G \approx 0.71070$ sits 0.5% above $1/\sqrt{2} \approx 0.70711$. The Landauer crossing proper lies between steps 35 and 36; the Shannon bound lies just above step 36 and below step 37.

Thirty-six is also the pentagonal angle: 36.00 degrees. The gap scaling ratio's leading term sits 0.36 degrees above the exact pentagon.

The distance to the next integer:

$$37 - 400/11 = 7/11$$

Boundary prime over corridor prime. The completion prime 37 stands exactly $7/11$ above the base ratio.

3. BINARY DRIFT SEQUENCE

Define:

$$\text{drift}(k) = \text{floor}(400/11 * 2^k) - 36 * 2^k$$

This measures how far the scaled ratio departs from pure 36-doubling. The fractional part $4/11$ accumulates under binary scaling, generating integer drift:

k = 0:	drift =	0	
k = 1:	drift =	0	
k = 2:	drift =	1	
k = 3:	drift =	2	
k = 4:	drift =	5	pentagonal prime
k = 5:	drift =	11	corridor prime
k = 6:	drift =	23	mirror point (18 + 5)
k = 7:	drift =	46	= 2 * 23
k = 8:	drift =	93	= 3 * 31
...			
k = 14:	drift =	5957	= 7 * 23 * 37

The framework's primes emerge in order:

5	appears at k = 4	($2^4 = 16$)
11	appears at k = 5	($2^5 = 32 = \text{Floor landmark}$)
23	appears at k = 6	($2^6 = 64 = \text{Pivot}$)

At $k = 14$ (where $14 = 2 * 7$ is the tower's breath span), the drift factorizes as:

$$5957 = 7 * 23 * 37$$

Boundary prime * mirror point * completion prime. Three framework numbers in one product, at the breath span's own doubling depth.

4. ERGODIC CYCLE MODULO 11

The fractional parts of $(400/11) * 2^k$ cycle through all residues modulo 11, because 2 is a primitive root of 11:

k = 0:	4/11	k = 5:	7/11
k = 1:	8/11	k = 6:	3/11
k = 2:	5/11	k = 7:	6/11
k = 3:	10/11	k = 8:	1/11
k = 4:	9/11	k = 9:	2/11
k = 10:	4/11	(cycle returns)	

Period = 10 = $\phi(11)$.

Binary doubling visits every possible state in the 11-world before returning. No residue is skipped. This is maximal exploration – full ergodic reach.

5. WHY 11 IS THE CORRIDOR PRIME

The primitive root status of 2 modulo each framework prime:

$\text{ord}(2 \bmod 3) = 2 = \phi(3)$	primitive root (interior prime)
$\text{ord}(2 \bmod 5) = 4 = \phi(5)$	primitive root (interior prime)
$\text{ord}(2 \bmod 7) = 3 \neq \phi(7)$	NOT primitive root
$\text{ord}(2 \bmod 11) = 10 = \phi(11)$	primitive root (corridor prime)
$\text{ord}(2 \bmod 13) = 12 = \phi(13)$	primitive root (intruder prime)

7 is the only framework prime where 2 is not a primitive root. Binary cannot fully explore the boundary – it reaches only 3 of 6 residue classes. This is consistent with 7's structural role: the boundary that locks rather than enables flow.

11 is the first prime outside the H4 closure $\{2, 3, 5\}$ and outside the boundary $\{7\}$ where binary still achieves full ergodic reach. It earns the name "corridor prime" because it is the first prime where binary has unrestricted passage.

6. INTEGER FLOORS – PRIMES ENTER IN ORDER

k = 0:	floor =	36 = $2^2 * 3^2$	(binary, structure)
k = 1:	floor =	72 = $2^3 * 3^2$	(binary deepens)
k = 2:	floor =	145 = $5 * 29$	(pentagonal enters)
k = 3:	floor =	290 = $2 * 5 * 29$	(doubles)

k = 4:	floor =	581 = 7 * 83	(boundary enters)
k = 5:	floor =	1163 = 1163 (prime)	(irreducible step)
k = 6:	floor =	2327 = 13 * 179	(intruder enters)

The prime factors of the integer floors follow the framework's prime entry sequence:

First:	2, 3	(binary, structure)	– already present in 36
Then:	5	(pentagonal)	– enters at k = 2
Then:	7	(boundary)	– enters at k = 4
Then:	13	(intruder)	– enters at k = 6

This is the same order in which primes organize atomic electron shells. Neither sequence was fitted to the other. Both emerge independently from their respective domains.

7. TOWER ALIGNMENT

The drift hits framework primes at tower landmarks:

k = 4:	drift = 5	at $2^4 = 16$	(below Floor)
k = 5:	drift = 11	at $2^5 = 32$	(the Floor: $32G \sim 1/\phi$)
k = 6:	drift = 23	at $2^6 = 64$	(the Pivot: $64G \sim \sqrt{\phi}$)

The doubling that reaches the Floor produces drift 11 (the corridor).
The doubling that reaches the Pivot produces drift 23 (the mirror).

23 = 18 + 5 = Romeo + pentagonal, and is the atomic number of Vanadium, the outbound mirror point in the tower's 18-14-18-14 breath pattern.

8. EVIDENCE CLASSIFICATION

Level 1 – Exact [Theorem]:

All drift values, factorizations, primitive root orders, and cycle periods are exact integer or modular computations. They can be verified with any calculator or CAS. Nothing is approximate.

Level 2 – Empirical [Observation]:

The alignment of prime emergence order with atomic shell structure and tower landmarks is an observed pattern. The pattern is precise (exact integers at exact positions) but the interpretation – that $400/11$ encodes the framework's prime hierarchy – is a claim about meaning, not a proof.

Level 3 – Conjectural [Conjecture]:

None claimed in this document. No fitting, no free parameters, no statistical arguments. The ratio $400/11$ was derived from the gap scaling formula. The doubling is the framework's own operation. The primes were already there.

Labels follow the canonical methodology page; this is a definitional alignment and does not change the scope, honesty, or calculations of this document.

9. FALSIFIABILITY

The drift sequence is deterministic. Any single factorization error falsifies the document. Specific checkable claims:

```
400/11 * 16384 = 6553600/11 = 595781.8181...
floor = 595781
595781 - 36 * 16384 = 595781 - 589824 = 5957
5957 / 7 = 851
851 / 23 = 37
Therefore 5957 = 7 * 23 * 37

ord(2 mod 7) = 3 (since 2^3 = 8 = 1 mod 7, but 2^1 = 2, 2^2 = 4)
ord(2 mod 11) = 10 (since 2^10 = 1024 = 1 mod 11)
```

Any computational tool can verify every claim in this document in under one second.

10. VERIFICATION

All arithmetic independently verified by:

- Exact rational computation (Python fractions module)
- Claude Opus (Anthropic), 3 April 2026
- Cross-checked against published gap scaling formula (Paper 4)

Key verification outputs:

drift(14) = 5957 = 7 * 23 * 37	CONFIRMED
ord(2 mod 7) = 3 $\neq$ 6 = phi(7)	CONFIRMED
ord(2 mod 11) = 10 = phi(11)	CONFIRMED
Ergodic cycle period = 10, visits all residues	CONFIRMED
Prime entry order: {2,3} -> 5 -> 7 -> 13	CONFIRMED

D. B. – April 2026
Gap Geometry
<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>
OSF: osf.io/zx4g7

K_AUD framework: November–December 2025
Binary scaling discovered: 3 April 2026
Documents prepared for publication: 3 April 2026
